

IRAKO, Japan

WMO: 476530

Lat: 34.6297N

Lon: 137.0939E

Elev: 8

StdP: 101.23

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.8	(c) 1.6	(d) -8.3	(e) 1.9	(f) 3.8	(g) -7.0	(h) 2.1	(i) 4.2	(j) 12.2	(k) 6.4	(l) 11.0	(m) 5.4	(n) 4.7	(o) 340	(p) 0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.3	(c) 33.0	(d) 25.8	(e) 31.7	(f) 25.6	(g) 30.6	(h) 25.3	(i) 27.1	(j) 30.4	(k) 26.6	(l) 29.7	(m) 26.1	(n) 29.1	(o) 4.1	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.3	21.8	28.8	25.7	21.0	28.4	25.2	20.4	28.0	85.6	30.4	83.3	29.8	81.1	29.3	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.2	(b) 9.1	(c) 8.2	(e) -1.3	(f) 35.2	(g) 0.9	(h) 1.3	(i) -1.9	(j) 36.1	(k) -2.4	(l) 36.9	(m) -2.9	(n) 37.6	(o) -3.5	(p) 38.5		
			DB													
			WB													
				(e) -3.2	(f) 27.7	(g) 1.0	(h) 0.8	(i) -3.8	(j) 28.3	(k) -4.4	(l) 28.8	(m) -4.9	(n) 29.3	(o) -5.6	(p) 29.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	6.0	6.4	9.4	14.4	19.0	22.5	26.6	28.1	24.8	19.4	13.6	8.5
	DBStd	7.94	1.96	2.33	2.63	2.75	1.96	2.06	2.15	1.73	2.40	2.51	2.68	2.49
	HDD10.0	336	126	103	43	2	0	0	0	0	0	0	2	59
	HDD18.3	1589	383	333	276	119	15	1	0	0	0	17	142	304
	CDD10.0	2754	1	3	26	135	281	374	514	559	443	293	111	14
	CDD18.3	965	0	0	0	2	37	124	256	301	193	51	1	0
	CDH23.3	7516	0	0	0	3	81	474	2278	3216	1350	113	0	0
	CDH26.7	2201	0	0	0	0	6	71	701	1136	282	6	0	0
(14) Wind	WSAvg	3.8	4.7	4.5	4.3	4.0	3.6	3.1	3.1	3.2	3.4	3.4	3.7	4.4
(15) Precipitation	PrecAvg	1627	61	75	116	140	168	186	158	128	253	179	110	61
	PrecMax	2206	187	193	199	265	364	325	333	313	502	639	219	152
	PrecMin	1040	1	22	49	26	70	65	11	14	65	30	9	6
	PrecStd	294	48	41	37	60	71	57	90	79	106	115	58	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	16.2	19.0	23.4	27.4	30.6	34.2	35.1	32.2	27.4	21.8	17.6
		MCWB	9.8	13.1	13.4	16.2	19.2	24.2	25.8	26.1	24.9	21.8	17.2	14.9
	2%	DB	11.6	13.6	17.3	21.7	25.3	28.5	32.7	33.4	30.5	25.7	20.3	15.1
		MCWB	7.8	9.6	13.1	16.2	18.4	23.0	25.7	26.0	24.4	20.9	16.3	11.7
	5%	DB	10.5	11.8	15.9	20.4	24.0	27.1	31.3	32.3	29.4	24.6	19.2	13.8
		MCWB	6.8	8.1	12.1	15.6	18.1	22.1	25.6	25.9	24.2	20.2	15.5	10.2
	10%	DB	9.4	10.4	14.4	19.2	22.9	25.9	30.2	31.2	28.3	23.4	18.1	12.6
		MCWB	5.9	6.9	10.6	14.9	17.7	21.4	25.3	25.7	23.8	19.2	14.6	9.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	14.4	16.1	20.0	22.3	25.5	27.7	28.0	26.7	24.2	19.3	15.7
		MCDB	12.6	15.6	17.0	20.7	24.1	28.4	31.5	31.7	29.6	25.5	20.2	17.0
	2%	WB	8.7	10.7	14.4	18.2	20.8	24.5	26.9	27.3	25.9	22.7	17.6	12.6
		MCDB	10.6	12.6	16.4	20.1	22.8	27.2	30.3	30.7	28.5	24.6	19.3	14.4
	5%	WB	7.5	8.9	12.8	17.0	20.0	23.6	26.3	26.9	25.4	21.3	16.4	11.0
		MCDB	9.7	11.0	15.1	19.2	22.3	26.1	29.4	30.0	27.9	23.6	18.4	13.3
	10%	WB	6.4	7.5	11.2	15.7	19.0	22.6	25.8	26.4	24.9	20.0	15.2	9.6
		MCDB	8.9	9.9	13.6	18.2	21.6	24.9	28.8	29.6	27.3	22.7	17.5	12.1
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.5	7.5	7.8	7.2	6.0	6.0	6.3	5.9	6.1	6.3	5.9
		MCDBR	7.3	8.6	9.9	9.5	9.2	7.7	7.7	7.6	6.8	6.8	7.3	7.4
		MCWBR	5.2	6.1	6.8	5.5	4.6	3.5	2.8	2.6	3.0	4.0	5.0	5.4
	5% WB	MCDBR	7.0	7.7	8.7	7.2	7.0	6.3	6.7	6.6	5.7	5.7	6.4	6.7
		MCWBR	5.6	6.5	7.4	5.4	4.6	3.2	2.8	2.6	2.9	4.4	5.4	5.9
(40) Clear-Sky Solar Irradiance	taub		0.387	0.430	0.490	0.534	0.541	0.563	0.556	0.528	0.492	0.465	0.417	0.386
	taud		2.220	2.112	1.960	1.902	1.955	1.973	2.042	2.110	2.179	2.171	2.217	2.238
	Ebn at Noon		794	798	780	767	767	748	750	764	772	755	755	768
	Edh at Noon		111	138	174	194	186	182	169	156	139	129	111	104
(44) All-Sky Solar Radiation	RadAvg		2.62	3.27	4.20	4.89	5.25	4.67	5.16	5.30	4.12	3.25	2.65	2.36
	RadStd		0.17	0.24	0.28	0.49	0.57	0.45	0.68	0.46	0.37	0.34	0.24	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	+0.42	+0.48	N/A	N/A	+84
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page